Model comparison

Wednesday, November 4, 2026

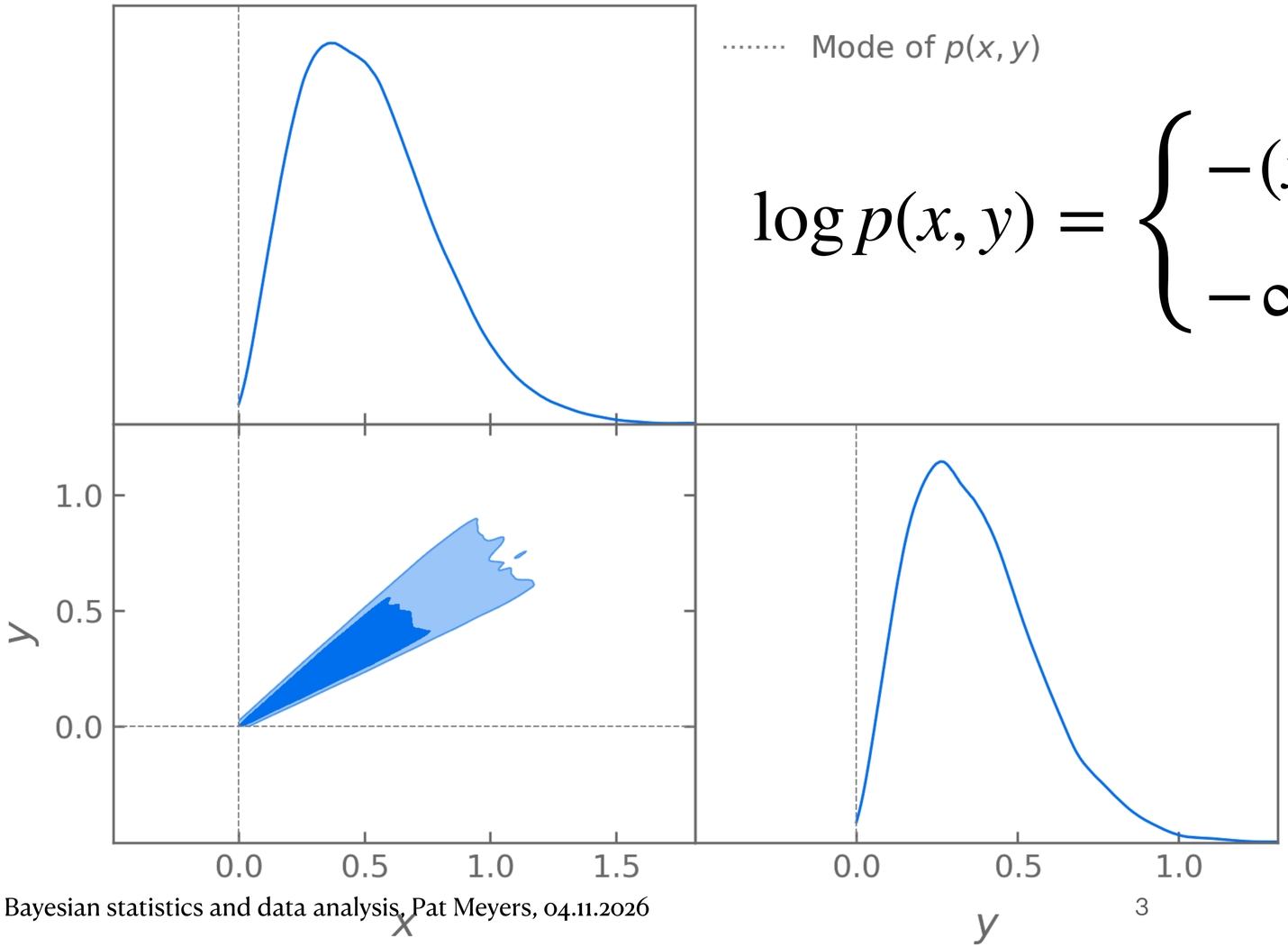
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Outline of the day

I've decided to split things up

- **This week:** Model selection and regression
  - Linear regression, generalized linear models
  - Brief rehash of last week
  - Recap: Marginal likelihoods, Bayes factors
  - Leave-one-out (LOO) cross-validation
  - Approximations/alternatives to LOO (WAIC, DIC, etc.)

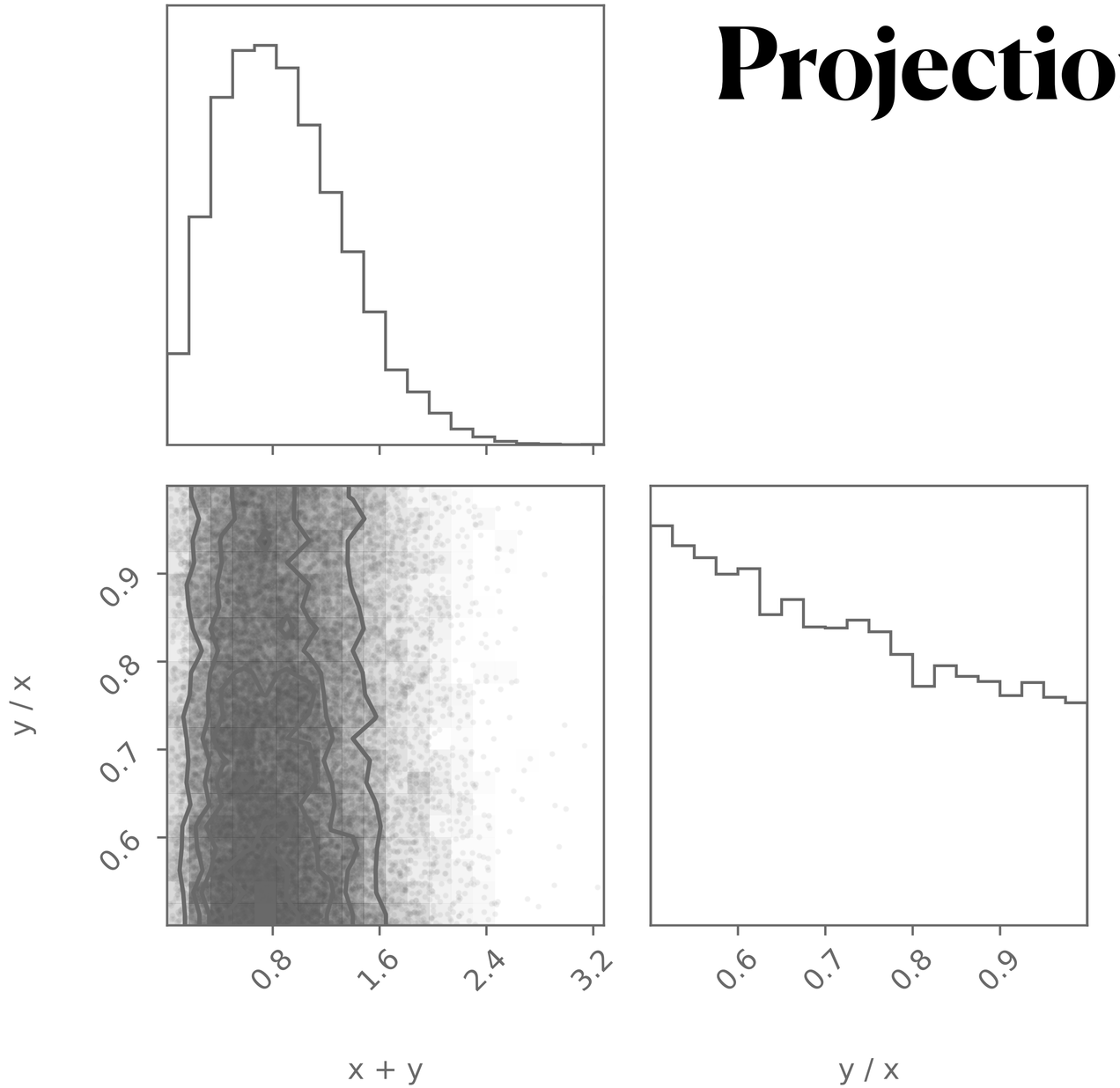
Projection effects — when marginals fail



$$\log p(x, y) = \begin{cases} -(x+y)^2 & \text{if } x, y \geq 0 \text{ and } \frac{x}{2} \leq y \leq x \\ -\infty & \text{otherwise.} \end{cases}$$

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Projection effects



- One way to deal with this is to look at combinations of the variables, e.g.
  - $s = x + y$ and $r = y/x$
  - **Also**, sometimes the marginals are telling you something!

Linear Regression

- We've already seen an example of this

$$y_i = mx_i + b = \begin{bmatrix} 1 & x_i \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix}$$

- We could upgrade this to be our quadratic fit, but it's still a linear model because it's **linear in the unknown parameters**
- $y_i = cx_i^2 + mx_i + b = \begin{bmatrix} 1 & x_i & x_i^2 \end{bmatrix} \begin{bmatrix} b \\ m \\ c \end{bmatrix}$

Linear Regression

- $y_i = cx_i^2 + mx_i + b = \begin{bmatrix} 1 & x_i & x_i^2 \end{bmatrix} \begin{bmatrix} b \\ m \\ c \end{bmatrix}$
- If, instead of writing out each equation for y_i separately, we write it in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_0 & x_0^2 \\ 1 & x_1 & x_1^2 \\ \vdots & \vdots & \vdots \end{bmatrix} \text{ and } \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

- Where we've now defined a set of **regressors** in our **design matrix** and their associated **coefficients.**

Linear Regression

- The columns of $\mathbf{X}$ correspond to data streams on which our data $\mathbf{y}$ depend. They don't have to be linear and quadratic in the same independent variable.
- We could decompose $\mathbf{y}$ in a Fourier series expansion. Then the columns of $\mathbf{X}$ would be sines and cosines at given frequencies, with rows corresponding to different times.

$$\mathbf{X} = \begin{pmatrix} 1 & \cos 2\pi f_0 t_0 & \sin 2\pi f_0 t_0 & \cos 2\pi f_1 t_0 & \sin 2\pi f_1 t_0 & \cdots \\ 1 & \cos 2\pi f_0 t_1 & \sin 2\pi f_0 t_1 & \cos 2\pi f_1 t_1 & \sin 2\pi f_1 t_1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Linear Regression

- The columns of $\mathbf{X}$ correspond to data streams on which our data $\mathbf{y}$ depend. They don't have to be linear and quadratic in the same independent variable.
- Or i could refer to patients in a medical study, and y_i some property we want to predict about that patient.
- Then the columns of $\mathbf{X}$ would correspond to different attributes of that patient, e.g. age, height, etc.

$$\mathbf{X} = \begin{pmatrix} 1 & a_0 & h_0 & \cdots \\ 1 & a_1 & h_1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Linear Regression

- In general we want to estimate β.
  - We might want to predict future $\mathbf{y}$ values reliably.
  - Or the β coefficients might be physical quantities we are trying to measure
- Let's assume the data $\mathbf{y}$ are Gaussian distributed:

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma) = \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$$

$$\text{equivalently } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \text{ with } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2\mathbf{I})$$

Linear Regression

- In this case, we can estimate the maximum likelihood coefficients **and** the covariance matrix on them (including estimating correlations)
- We can take derivatives of our likelihood w.r.t. β and set it equal to zero and find:

$$\hat{\boldsymbol{\beta}}_{MLE} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

- Likewise, we can estimate $\boldsymbol{\Sigma}_{\hat{\beta}} = \mathrm{E}\left[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T\right]$:

$$\boldsymbol{\Sigma}_{\hat{\beta}} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1},$$

- This assumes the data are **uncorrelated and all have the same variance**.

Linear Regression

- We can relax this assumption by upgrading from

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma) = \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$$

$$\text{to } p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \boldsymbol{\Sigma}) = \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma})$$

- So our data can now have any general correlation it wants. In this case we revise the previous MLE.
- We can see “inverse noise weighting” firsthand here!

$$\hat{\boldsymbol{\beta}}_{MLE} = (\mathbf{X}^T \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \Sigma^{-1} \mathbf{y}\,,$$

$$\boldsymbol{\Sigma}_{\hat{\beta}} = (\mathbf{X}^T \Sigma^{-1} \mathbf{X})^{-1}.$$

- What do I need to make this Bayesian?

Linear Regression

- Priors!
- So far we have just looked at likelihoods. Now let's look at priors.

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \boldsymbol{\Sigma}) \propto p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \boldsymbol{\Sigma}) p(\boldsymbol{\beta})$$

- We can use any sampler we'd like to now evaluate this once we've decided on a prior.
- If we use a Gaussian prior on $\boldsymbol{\beta}$, e.g. $p(\boldsymbol{\beta}) = \mathcal{N}(0, \boldsymbol{\varphi})$, then actually the posterior itself is analytic and we end up with a Gaussian posterior:

Linear Regression

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$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \boldsymbol{\Sigma}, \boldsymbol{\varphi}) = \mathcal{N}_{\beta}(\boldsymbol{\mu}_{\beta}, \Sigma_{\beta})$$

- Where I've used the subscript $\mathcal{N}_{\beta}$ to indicate that the normal distribution is on β instead of $\mathbf{y}$.

Linear Regression

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$$\boldsymbol{\mu}_\beta = (\mathbf{X}^T \Sigma^{-1} \mathbf{X} + \boldsymbol{\varphi}^{-1})^{-1} \mathbf{X}^T \Sigma^{-1} \mathbf{y}$$

$$\boldsymbol{\Sigma}_{\boldsymbol{\beta}} = (\mathbf{X}^T \Sigma^{-1} \mathbf{X} + \boldsymbol{\varphi}^{-1})^{-1}$$

Linear Regression

- If we use a Gaussian prior on $\boldsymbol{\beta}$, e.g. $p(\boldsymbol{\beta}) = \mathcal{N}(0,\boldsymbol{\varphi})$, then actually the posterior itself is analytic and we end up with a Gaussian posterior:

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$$\boldsymbol{\mu}_\beta = (\mathbf{X}^T\Sigma^{-1}\mathbf{X} + \boldsymbol{\varphi}^{-1})^{-1}\mathbf{X}^T\Sigma^{-1}\mathbf{y}$$

$$\boldsymbol{\Sigma}_{\boldsymbol{\beta}} = (\mathbf{X}^T\Sigma^{-1}\mathbf{X} + \boldsymbol{\varphi}^{-1})^{-1}$$

- So if your priors are Gaussian, then you don't even need to sample!
- You can often approximate the behavior of a uniform prior with a wide Gaussian prior (except enforcing things like positivity).
- This is safest when you know the data will be informative.

Linear Regression

- Let's think about these priors.
- What does it mean if, say $\varphi = \tau^2\mathbf{I}$? ($\tau$ is not a vector, it's a scalar here)
- What if we think that all of our coefficients come from the same distribution...but we're not sure about that distribution?
- We can make τ a free parameter, and it parameterizes *the prior,* so we call it a **hyperparameter.**

Linear Regression

- We can make τ a free parameter, and it parameterizes *the prior,* so we call it a **hyperparameter.**
- We now have a hierarchical model:
  - **Likelihood**: $\mathbf{y} \mid \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma})$
  - **Prior**: $\boldsymbol{\beta} \mid \tau \sim \mathcal{N}_{\beta}\left[0, \boldsymbol{\varphi}(\tau)\right]$
  - **Hyperprior**: $\tau \sim p(\tau)$
- We can sample all of our free parameters at once if we wanted to!
- **Or** in some cases we can even choose $p(\tau)$ such that things remain analytic using a **conjugate prior**.

Linear Regression

- In some cases, we are not interested in $\boldsymbol{\beta}$; they are "nuisance parameters"
- Instead we're interested in τ (or whatever parameters are used to create $\boldsymbol{\varphi}$, call them $\boldsymbol{\theta}$)
- In this case, because we have a Gaussian likelihood and a Gaussian prior, we can analytically marginalize over $\boldsymbol{\beta}$:

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}, \boldsymbol{\Sigma}) \propto p(\boldsymbol{\theta}) \int \mathcal{N}_y(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}) \mathcal{N}_\beta(0, \boldsymbol{\varphi}(\boldsymbol{\theta}))\, d\boldsymbol{\beta}$$

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}, \boldsymbol{\Sigma}) \propto p(\boldsymbol{\theta}) \mathcal{N}_y \left[0, \mathbf{X}\boldsymbol{\varphi}(\boldsymbol{\theta})\mathbf{X}^T + \boldsymbol{\Sigma}\right] .$$

- Gaussians are wonderful this way. And it's my pleasure to point you to the **matrix cookbook**, which has lots of nice identities worked out for working with them.

Linear Regression

$$p(\boldsymbol{\theta} \mid \mathbf{y}, \mathbf{X}, \boldsymbol{\Sigma}) \propto p(\boldsymbol{\theta}) \mathcal{N}_y \left[0, \mathbf{X}\boldsymbol{\varphi}(\boldsymbol{\theta})\mathbf{X}^T + \boldsymbol{\Sigma}\right] \ .$$

- This thing has the form of a **Gaussian process** model for our data $\mathbf{y}$, which we'll return to later in the course.
- It's not really a coincidence that we end up here from where we started, and when we come back to discussing Gaussian processes this will be a nice derivation to keep in mind.

Generalized linear models

- We can generalize this idea to **Generalized linear models:**
  - Instead of $\mathbf{X}\boldsymbol{\beta}$ estimating the mean of a distribution, let's let it estimate some **function of the mean.**
    - $g(\boldsymbol{\mu}) = \mathbf{X}\boldsymbol{\beta}$, where $\boldsymbol{\mu} = \mathrm{E}[y \mid \mathbf{X}, \boldsymbol{\beta}]$.
  - And while we're at it, let's not just use Gaussians!

Generalized linear models

- An example:
  - Let y_i represent Poisson counts, and μ_i be the expected number of counts in that bin.
  - Define our "link function" $g(\mu_i) = \ln \mu_i = (\mathbf{X}\boldsymbol{\beta})_i$.
- Our Poisson likelihood is then:

$$p(y_i \,|\, \mu_i) = \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!}$$

- and we can substitute in for our model:

$$p(y_i \,|\, \mathbf{X}, \boldsymbol{\beta}) = \frac{e^{y_i (\mathbf{X}\boldsymbol{\beta})_i} \exp\left[-e^{(\mathbf{X}\boldsymbol{\beta})_i}\right]}{y_i!}\,.$$

And now for something completely different...

Model comparison

- Two common ways to think about this:
  - Comparing the **fits** of different models (relative to the flexibility of the model)
  - Comparing the **predictive power** of different models
- We will discuss both
- Both can be useful and will produce other interesting data products along the way

Bayes factors and Odds ratios

- Discussed this a few weeks ago in the context of nested sampling.
- Bayes' theorem gives us a way to write down posterior probabilities on models that we want to compare:

$$\mathcal{O}_{M_A}^{M_B} = \frac{p(M_B)}{p(M_A)} \frac{p(d\,|\,M_B)}{p(d\,|\,M_A)}$$

- **First term**: prior **odds.** Do we favor model B more than A before we start?
- **Second term:** *"Bayes factor"* ($\mathcal{B}_{M_A}^{M_B}$)— which model does the data prefer?
- If these are the only two choices, and one of the models is "correct" then the Odds ratio has the interpretation of being "betting odds" between the two models.

Bayes factors — Laplace approximation

- Each term in the Bayes factor is the **Marginal Likelihood (**sometimes called the “evidence”)

$$p(d \mid M_A) = \int d\vec{\theta}\, p(d \mid \vec{\theta}, M_A) p(\vec{\theta} \mid M_A)\,.$$

- We can approximate this integral analytically if the data are informative using the “Laplace approximation”

$$p(d \mid M) \approx p(d \mid \theta_{MLE}, M) \frac{\Delta\mathscr{V}}{\mathscr{V}}$$

- Where $\Delta\mathscr{V}$ is the spread of the likelihood function, and $\mathscr{V}$ is the spread of the prior.
- This ratio $\Delta\mathscr{V}/\mathscr{V}$ is called the “**Occam penalty**”.

Bayes factors — Laplace approximation

- If $\Delta\mathscr{V} \approx \mathscr{V}$ then the likelihood is uninformative, and the marginal likelihood is given approximately by the max likelihood.
- **Question**:
  - If $\Delta\mathscr{V}_A/\mathscr{V}_A < \Delta\mathscr{V}_B/\mathscr{V}_B$, but the **likelihoods are the same for the two models** then which model does the Bayes factor prefer?

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- **Answer:**
  - It prefers model **B.**

$$\frac{p(d\,|\,M_B)}{p(d\,|\,M_A)} \approx \frac{p(d\,|\,\vec{\theta}_{MLE})}{p(d\,|\,\vec{\theta}_{MLE})}\frac{\Delta\mathscr{V}_B/\mathscr{V}_B}{\Delta\mathscr{V}_A/\mathscr{V}_A} = \frac{\Delta\mathscr{V}_B/\mathscr{V}_B}{\Delta\mathscr{V}_A/\mathscr{V}_A} > 1.$$

Bayes factors — Laplace approximation

- If we have two similar likelihoods but the priors for one model are much broader than those of the other, we prefer the **simpler model.**
  - If we have extra parameters in one model, but the fit is not better, Bayes factors naturally prefer the simpler model
  - If we have the same likelihood and two different priors, **then if the priors do not change $\Delta\mathscr{V}$ (i.e. they don't add information)** we will prefer the more restrictive prior

Bayes factors — consequences

- If your priors are arbitrary, then Bayes factors can sometimes not be helpful
- If your priors are improper (e.g. if a value can be anywhere on the real line) then the Bayes factors are meaningless!!

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Bayes factors — consequences

- If your priors are arbitrary, then Bayes factors can sometimes not be helpful
- If your priors are improper (e.g. if a value can be anywhere on the real line) then the Bayes factors are meaningless!!
- **Question:** If the priors are the same, but the likelihoods are different, what is the Bayes factor testing?
- **Answer:** Both how well the model fits **and** the spread of the likelihood. $\mathscr{V}$ is the same, so we have

$$\mathscr{B}^{M_B}_{M_A} \approx \frac{p(d\,|\,\hat{\theta}_B, M_B)}{p(d\,|\,\hat{\theta}_A, M_A)} \frac{\Delta\mathscr{V}_B}{\Delta\mathscr{V}_A}$$

Bayes factors

- In modern statistics books, Bayes factors and Odds ratios are often played down and not used, in favor of **predictive checks** or **model exploration** (what we've been doing so far).
- However, they do have uses—e.g. if there are well-defined "signal" and "noise" models.

Bayes factors — how to calculate them

- Calculating Bayes factors can be difficult (we saw this during the nested sampling lecture)
- Usually we need to calculate the **marginal likelihood** for both models which is challenging, especially in high dimensions. Simple means of the likelihood or harmonic means don't cut it.
  - **Nested sampling**
  - **Parallel tempering** + **"thermodynamic integration" or "stepping stone"**
  - **Modern machine learning techniques**

Bayes factors — how to calculate them

- We can also do direct model comparison
  - If the models are nested, we can use **Savage-Dickey Bayes factors**
  - Bridge sampling
  - **"Product space"** sampling (each model has an indicator variable, sample the indicators)
  - **Importance sampling** between models (usually requires posterior distributions to be similar)

Savage-Dickey Bayes factors

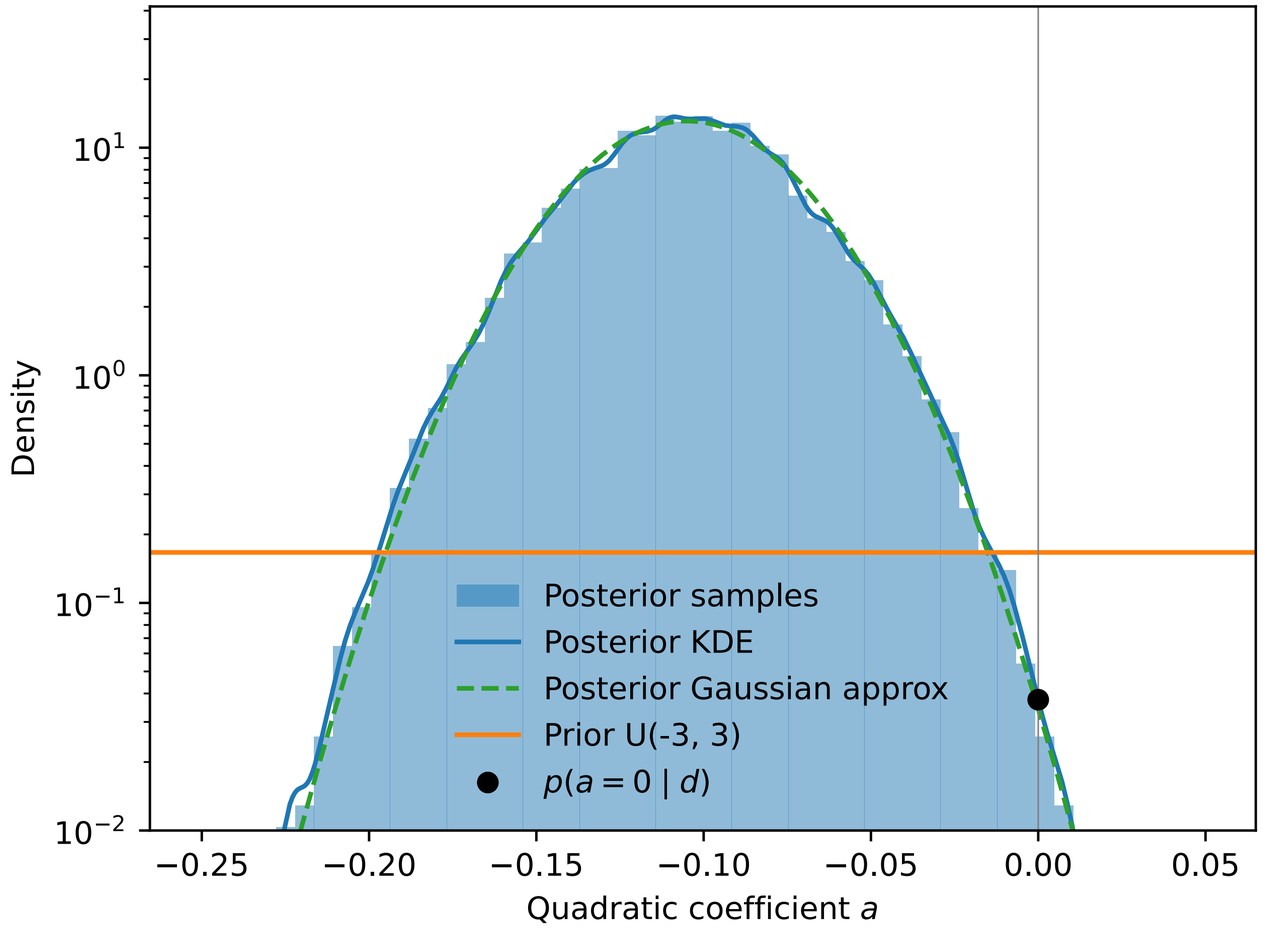
- In situations where we are comparing two **nested models,** we can calculate a Savage-Dickey Bayes factor.
- For example, if our **quadratic** model is
- $f(x) = mx + cx^2 + b$,
- then when $c = 0$, we reduce to a simple **linear** model. In this situation, we can calculate the Savage-Dickey Bayes factor between these two models:

$$\mathcal{B}^{quad}_{lin} = \frac{p(c = 0)}{p(c = 0 \mid d)}.$$

Savage-Dickey Bayes factors

$$\mathcal{B}^{quad}_{lin} = \frac{p(c=0)}{p(c=0\,|\,d)}\,.$$

- We can do this because the models are nested. **Bonus exercise: derive this formula.**
- For the linear model we've been using, we can compare the posterior and the prior.
- This can come in handy for comparing hierarchical models, which can often be nested.



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Beyond Bayes factors — predictive tests

- Different paradigm: **which model is better at predicting new data?**
- **Gold standard** — "Cross-validation"
  - Hold out some data, fit a model on what's left, evaluate fit on the left-out data using **posterior predictive likelihood.**
  - Previously, we've simulated from the posterior predictive, now we're using it to evaluate model fit.

Cross-validation

- Let $\vec{y}$ be our data, and $\vec{y}_k \equiv \{y_j\}_{j \in \mathscr{K}}$ where $\text{size}(\mathscr{K}) = k$.
- Then $\vec{y}_{-k} = \{y_j\}_{j \notin \mathscr{K}}$.
- Then our posterior predictive likelihood is:

$$p(\vec{y}_k \mid \vec{y}_{-k}) = \int p(\vec{y}_k \mid \vec{\theta}) p(\vec{\theta} \mid \vec{y}_{-k}) \, \mathrm{d}\vec{\theta}$$

- In words: we average the likelihood on the left-out data **over** the posterior on the left-in data.

Cross-validation

- We can define a “Cross-validation score” that averages over all possible subsets we could draw

$$S_{CV}(k) = \binom{n}{k}^{-1} \sum_{|V|=k} \frac{1}{k} \sum_{j \in V} \ln p(\vec{y}_j \,|\, \vec{y}_{-V})$$

- Where “V” is one draw of size “k” from our “n” data points.
- If we have multiple models, we could look at the ratio of the scores.

Cross-validation

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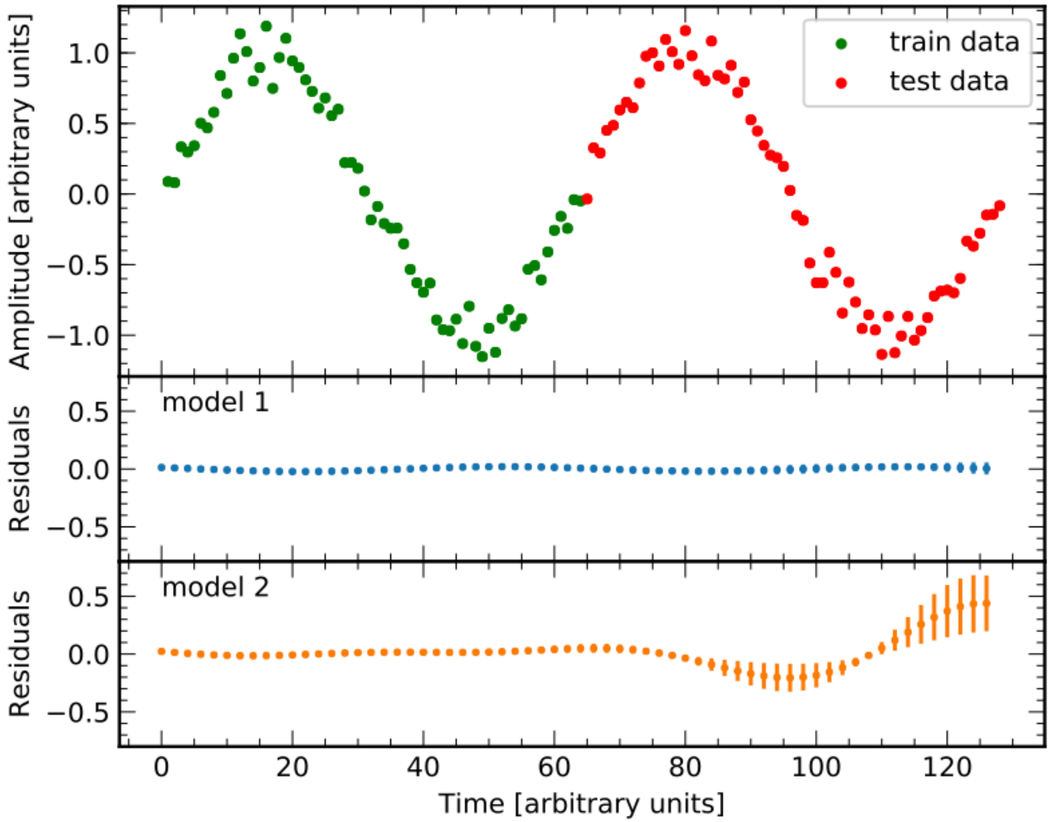
$$S_{CV}(k) = \binom{n}{k}^{-1} \sum_{|V|=k} \frac{1}{k} \sum_{j \in V} \ln p(y_j \,|\, \vec{y}_{-V})$$

- An interesting identity noted in Fong & Holmes 2020 links the marginal likelihood to these scores:

$$\ln p(d \,|\, M) = \sum_{k=1}^{n} S_{CV}(k)$$

Gaining some intuition

- Cross-validation penalizes overfitting, but in a different way than a Bayes factor
  - If we leave out data and we fit to what's left and we are too confident in our fit, then we will have a poor prediction.
  - This effect grows as k grows, because we rely more and more on the prior, for example.



Wang, Taylor, Vallisneri 2019

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Leave-one-out cross-validation (LOO-CV)

- Often, we will calculate $S_{CV}(1)$, that is — "leave-one-out" (**LOO**) cross-validation (CV). We leave out just a single data point each time, which we'll call i.
- We therefore define the "log point-wise predictive density"

$$\begin{aligned}\text{lppd}_{\text{loo-cv}} \equiv \sum_{i=1}^{n} \ln p(y_i \,|\, y_{-i}) &= \sum_{i=1}^{n} \ln \left[\int p(y_i \mid \theta) p(\theta \mid y_{-i}) \, d\theta \right] \\ &\approx \sum_{i=1}^{n} \ln \left[\frac{1}{S} \sum_{s=1}^{S} p(y_i \,|\, \theta^s) \right]\end{aligned}$$

- Where in the last step we are doing a Monte Carlo integral $\theta^s \sim p(\vec{\theta} \,|\, \vec{y}_{-i})$.

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LOO-CV

- If we want to compare two models, we can either compare the total $\text{lppd}_{\text{loo-cv}}$ for each model
- **OR** we could compare them point-wise between the models. That is:

$$D_i = \text{lppd}_{i,M1} - \text{lppd}_{i,M2}$$

- And compare the mean, $\bar{D}$ and its standard error, $\sigma_D/\sqrt{n}$.
- If one model is favored over the other, $\bar{D}$ will be shifted from zero and $\sigma_D/\sqrt{n}$ tells us the uncertainty on that value.

LOO-CV

- This does mean refitting the model *n* times! If the model takes a long time to fit, then that's an issue.
- There are ways around this, known as "Importance sampling" (where one can repurpose posterior samples from one fit to use for a second fit) and "Pareto-smoothed importance sampling" (PSIS), which helps deal with complexities of typical importance sampling.
- As part of PSIS, one often gets a statistic out that identifies whether a data point is particularly important, or an outlier.
- I won't discuss these in detail, but **arviz**'s "**loo()**" function uses this method by default.

Approximations or replacements for LOO-CV

WAIC

- Because LOO-CV can be cumbersome to compute, people often use alternatives.
- An asymptotic approximation to LOO-CV is the “Widely Applicable Information Criterion.”
- We don’t refit the model each time
- We predict individual data points using a single fit with all of the data, and use a correction term because this will be overconfident.

WAIC

- Defined in **Watanabe 2010**. We first break up our data into n batches
- We need the likelihood to factorize on these batches. We then calculate

$$\text{lppd} \equiv \sum_{i=1}^{n} \ln p(y_i \,|\, y) = \sum_{i=1}^{n} \ln \left[\int p(y_i \mid \theta) p(\theta \mid y) \, d\theta \right]$$

- We correct this to account for the fact that the red term is now y instead of y_{-i}, which makes us overconfident.

$$\text{WAIC} = -2[\widehat{\text{elpd}}_{\text{WAIC}}] = -2\left(\text{lppd} - p_{WAIC}\right)$$

where "elpd" is the "expected log point wise predictive density" (i.e. it includes the correction)

$$p_{WAIC} = \sum_{i=1}^{n} \text{var}_s \left[\ln p(y_i \,|\, \theta^s) \right] \, .$$

- That is, we take the variance of the log-likelihood on a specific point **over the posterior samples**. We then sum up those variances.

WAIC

- It turns out that this p_{WAIC} can be interpreted as the **number of parameters in the model that are constrained by the data.**
- So if we compare WAICs calculated on two models, then we have recovered something like an overfitting penalty or an Occam penalty.

WAIC in words

1. Fit the model using all of the data to get a list of posterior samples, $\{\theta^s\}_{s=1}^S \sim p(\theta \mid y)$.
2. Partition the data into n chunks.
3. Calculate the lppd
    1. For each chunk, calculate the likelihood on just that chunk over all samples $\{p(y_i \mid \theta^s)\}_{s=1}^S$
    2. Average across samples, take the log, to approximate $\ln p(y_i \mid y)$
    3. Sum across the log predictive densities
4. Calculate p_{WAIC}
    1. Using $\{\ln p(y_i \mid \theta^s)\}_{s=1}^S$, calculate the variance for each i across samples.
    2. Sum the variances

When WAIC doesn't work

- WAIC requires our likelihood to factorize in some way so that we don't have to refit the model to a limited amount of data.
- This doesn't always work in practice.
- In those cases, there are other potential criteria.
  - The Deviance Information Criterion
  - The Bayesian Information Criterion
  - The Akaike Information Criterion

Deviance Information Criterion (DIC)

- "Somewhat Bayesian"

$$DIC = -2\left[\ln p(y \mid \hat{\theta}_{Bayes}) - p_{DIC}\right]$$

- Where we have a different attempt at quantifying the **effective** number of constrained parameters

$$p_{DIC} = 2\left(\ln p(y \mid \hat{\theta}_{Bayes}) - \frac{1}{S}\sum_{s=1}^{S} \ln\left[p(y \mid \theta^{(s)})\right]\right)$$

- Where $\hat{\theta}_{Bayes}$ is the **posterior mean**.

Akaike Information Criterion (AIC)

- $AIC = -2\left[\ln p(y \mid \hat{\theta}_{MLE}) - k\right]$
- Where k is the number of parameters in the model.
- This is derived from the special case of a Gaussian, linear model with known variance and uniform priors.
- When this isn't satisfied, k can be different from, e.g. the p_{WAIC} or p_{DIC} which try to account for cases where the likelihood is not informative about certain parameters.

Summary of Information Criteria

Criterion	Definition	Penalty	Notes
AIC	$-2\big(\ln p(y \mid \hat{\theta}_{\mathrm{MLE}}) - k\big)$	k	MLE plug-in; regular models, large N
DIC	$-2\big(\ln p(y \mid \hat{\theta}_{\mathrm{Bayes}}) - p_{\mathrm{DIC}}\big)$	$p_{\mathrm{DIC}} = 2\Big(\ln p(y \mid \hat{\theta}_{\mathrm{Bayes}}) - \frac{1}{S}\sum_s \ln p(y \mid \theta^{(s)})\Big)$	Not reparam.-invariant; p_{DIC} can be negative
WAIC	$-2\big(\mathrm{lppd} - p_{\mathrm{WAIC}}\big)$ $\mathrm{lppd} = \sum_i \ln\Big(\frac{1}{S}\sum_s p(y_i \mid \theta^{(s)})\Big)$	$p_{\mathrm{WAIC}} = \sum_i \mathrm{V}_s\big[\ln p(y_i \mid \theta^{(s)})\big]$	Needs conditionally independent y_i $\approx$ LOO asymptotically
LOO-CV	$-2\sum_i \ln p(y_i \mid y_{-i})$ $p(y_i \mid y_{-i}) \approx \frac{1}{S}\sum_s p(y_i \mid y_{-i}, \theta^{(s)}_{-i})$	implicit: $p_{\mathrm{loo}} = \mathrm{lppd} - \widehat{\mathrm{elpd}}_{\mathrm{loo}}$	Correlated data allowed; PSIS avoids refitting